Graphs 6 — Cubic graphs · Reciprocal graphs

Calculator allowed; graph paper needed. Marks in brackets. ★ = harder question. Plot the points and join them with a *smooth* curve — never with a ruler.

1 **a** Copy and complete the table of values for $y = x^3 - 3x^2 - x + 3$, then draw the graph for $-2 \le x \le 4$. **b** Write down the values of x where the graph cuts the x-axis. [6]

x	-2	-1	0	1	2	3	4
y	-15		3		-3		15

2 The depth of water D metres in a tidal pool, t hours after 6 a.m., is given by $D = t^3 - 6t^2 + 9t + 2$ for $0 \le t \le 4$. [9]

a Copy and complete the table and draw the graph of D against t.

b Use your graph to estimate the greatest depth and the time when it occurs.

c Between which times is the depth less than 3 m?

t (hours)	0	1	2	3	4
D (m)	2		4		6

3 The cost £y per item of producing x hundred items is modelled by $y = \dfrac{x}{2} + \dfrac{8}{x}$ for $1 \le x \le 8$. [10]

a Copy and complete the table (2 d.p.).

b Draw the graph for $1 \le x \le 8$, with y from 0 to 9.

c Use your graph to estimate the cost per item when 350 items are made.

d How many items should be made for the lowest cost per item, and what is that cost?

x	1	2	3	4	5	6	8
$\frac{x}{2}$	0.5		1.5		2.5		4
$\frac{8}{x}$	8		2.67		1.6		1
y	8.50		4.17		4.10		5.00

Total: 25 marks

Check: build a table row by row (x^3, $-3x^2$, …) and add · cubes of negatives are negative · smooth curve through every point, no straight segments · a reciprocal graph never touches $x = 0$ · read a graph by drawing a horizontal or vertical line and marking the crossing.